JavaScript is disabled on your browser.

Java SE 27 & JDK 27

Class DataOutputStream

java.lang.Object
  java.io.OutputStream
    java.io.FilterOutputStream
      java.io.DataOutputStream

All Implemented Interfaces:

Closeable, DataOutput, Flushable, AutoCloseable

```
public class DataOutputStream
extends FilterOutputStream
implements DataOutput
```

A data output stream lets an application write primitive Java data types to an output stream in a portable way. An application can then use a data input stream to read the data back in. A data output stream wraps another output stream and delegates writing bytes to the write methods of that output stream. Writing data consisting of more than a single byte may cause several writes to the underlying output stream.

A DataOutputStream is not safe for use by multiple concurrent threads. If a DataOutputStream is to be used by more than one thread then access to the data output stream should be controlled by appropriate synchronization.

Since:

1.0

See Also:

DataInputStream

Field Summary

Fields

Modifier and Type	Field	Description
protected int	written	The number of bytes written to the data output stream so far.

Fields declared in class FilterOutputStream

out

Constructor Summary

Constructors

Constructor	Description
`DataOutputStream (OutputStream out)`	Creates a new data output stream to write data to the specified underlying output stream.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
`void`	`flush()`	Flushes this data output stream.
`final int`	`size()`	Returns the current value of the counter `written`, the number of bytes written to this data output stream so far.
`void`	`write(byte[] b, int off, int len)`	Writes `len` bytes from the specified byte array starting at offset `off` to the underlying output stream.
`void`	`write(int b)`	Writes the specified byte (the low eight bits of the argument `b`) to the underlying output stream.
`final void`	`writeBoolean(boolean v)`	Writes a `boolean` to the underlying output stream as a 1-byte value.
`final void`	`writeByte(int v)`	Writes out a `byte` to the underlying output stream as a 1-byte value.
`final void`	`writeBytes(String s)`	Writes out the string to the underlying output

		stream as a sequence of bytes.
final void	**writeChar**(int v)	Writes a char to the underlying output stream as a 2-byte value, high byte first.
final void	**writeChars**(String s)	Writes a string to the underlying output stream as a sequence of characters.
final void	**writeDouble**(double v)	Converts the double argument to a long using the doubleToLongBits method in class Double, and then writes that long value to the underlying output stream as an 8-byte quantity, high byte first.
final void	**writeFloat**(float v)	Converts the float argument to an int using the floatToIntBits method in class Float, and then writes that int value to the underlying output stream as a 4-byte quantity, high byte first.
final void	**writeInt**(int v)	Writes an int to the underlying output stream as four bytes, high byte first.
final void	**writeLong**(long v)	Writes a long to the underlying output stream as eight bytes, high byte first.

final void	**writeShort**(int v)	Writes a short to the underlying output stream as two bytes, high byte first.
final void	**writeUTF**(String str)	Writes a string to the underlying output stream using modified UTF-8 encoding in a machine-independent manner.

Methods declared in class FilterOutputStream

close, write

Methods declared in class OutputStream

nullOutputStream

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Methods declared in interface DataOutput

write

Field Details

written

```
protected int written
```

The number of bytes written to the data output stream so far. If this counter overflows, it will be wrapped to Integer.MAX_VALUE.

Constructor Details

DataOutputStream

```
public DataOutputStream(OutputStream out)
```

Creates a new data output stream to write data to the specified underlying output stream. The counter `written` is set to zero.

Parameters:

`out` - the underlying output stream, to be saved for later use.

See Also:

`FilterOutputStream.out`

Method Details

write

```
public void write(int b)
           throws IOException
```

Writes the specified byte (the low eight bits of the argument `b`) to the underlying output stream. If no exception is thrown, the counter `written` is incremented by `1`.

Implements the `write` method of `OutputStream`.

Specified by:

`write` in interface `DataOutput`

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the `byte` to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

write

```
public void write(byte[] b,
                  int off,
                  int len)
           throws IOException
```

Writes `len` bytes from the specified byte array starting at offset `off` to the underlying output stream. If no exception is thrown, the counter `written` is incremented by `len`.

Specified by:

`write` in interface `DataOutput`

Overrides:

`write` in class `FilterOutputStream`

Parameters:

`b` - the data.

`off` - the start offset in the data.

`len` - the number of bytes to write.

Throws:

`IOException` - if an I/O error occurs.

`IndexOutOfBoundsException` - If `off` is negative, `len` is negative, or `len` is greater than `b.length - off`

See Also:

`FilterOutputStream.out`

flush

```
public void flush()
           throws IOException
```

Flushes this data output stream. This forces any buffered output bytes to be written out to the stream.

The `flush` method of `DataOutputStream` calls the `flush` method of its underlying output stream.

Specified by:

`flush` in interface `Flushable`

Overrides:

`flush` in class `FilterOutputStream`

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`, `OutputStream.flush()`

writeBoolean

```
public final void writeBoolean(boolean v)
                        throws IOException
```

Writes a `boolean` to the underlying output stream as a 1-byte value. The value `true` is written out as the value `(byte)1`; the value `false` is written out as the value `(byte)0`. If no exception is thrown, the counter `written` is incremented by 1.

Specified by:

`writeBoolean` in interface `DataOutput`

Parameters:

`v` - a `boolean` value to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeByte

```
public final void writeByte(int v)
                     throws IOException
```

Writes out a `byte` to the underlying output stream as a 1-byte value. If no exception is thrown, the counter `written` is incremented by `1`.

Specified by:

`writeByte` in interface `DataOutput`

Parameters:

`v` - a `byte` value to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeShort

```
public final void writeShort(int v)
                      throws IOException
```

Writes a `short` to the underlying output stream as two bytes, high byte first. If no exception is thrown, the counter `written` is incremented by `2`.

Specified by:

`writeShort` in interface `DataOutput`

Parameters:

`v` - a `short` to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeChar

```
public final void writeChar(int v)
                     throws IOException
```

Writes a `char` to the underlying output stream as a 2-byte value, high byte first. If no exception is thrown, the counter `written` is incremented by 2.

Specified by:

`writeChar` in interface `DataOutput`

Parameters:

`v` - a `char` value to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeInt

```
public final void writeInt(int v)
                    throws IOException
```

Writes an `int` to the underlying output stream as four bytes, high byte first. If no exception is thrown, the counter `written` is incremented by 4.

Specified by:

`writeInt` in interface `DataOutput`

Parameters:

`v` - an `int` to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeLong

```
public final void writeLong(long v)
                     throws IOException
```

Writes a `long` to the underlying output stream as eight bytes, high byte first. In no exception is thrown, the counter `written` is incremented by 8.

Specified by:

`writeLong` in interface `DataOutput`

Parameters:

v - a `long` to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`

writeFloat

```
public final void writeFloat(float v)
                      throws IOException
```

Converts the float argument to an `int` using the `floatToIntBits` method in class `Float`, and then writes that `int` value to the underlying output stream as a 4-byte quantity, high byte first. If no exception is thrown, the counter `written` is incremented by 4.

Specified by:

`writeFloat` in interface `DataOutput`

Parameters:

v - a `float` value to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

`FilterOutputStream.out`, `Float.floatToIntBits(float)`

writeDouble

```
public final void writeDouble(double v)
                       throws IOException
```

Converts the double argument to a `long` using the `doubleToLongBits` method in class `Double`, and then writes that `long` value to the underlying output stream as an 8-byte quantity, high byte first. If no exception is thrown, the counter `written` is incremented by 8.

Specified by:

`writeDouble` in interface `DataOutput`

Parameters:

v - a `double` value to be written.

Throws:

`IOException` - if an I/O error occurs.

See Also:

FilterOutputStream.out,
Double.doubleToLongBits(double)

writeBytes

```
public final void writeBytes(String s)
                      throws IOException
```

Writes out the string to the underlying output stream as a sequence of bytes. Each character in the string is written out, in sequence, by discarding its high eight bits. If no exception is thrown, the counter `written` is incremented by the length of `s`.

Specified by:

writeBytes in interface DataOutput

Parameters:

`s` - a string of bytes to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

FilterOutputStream.out

writeChars

```
public final void writeChars(String s)
                      throws IOException
```

Writes a string to the underlying output stream as a sequence of characters. Each character is written to the data output stream as if by the `writeChar` method. If no exception is thrown, the counter `written` is incremented by twice the length of `s`.

Specified by:

writeChars in interface DataOutput

Parameters:

`s` - a `String` value to be written.

Throws:

IOException - if an I/O error occurs.

See Also:

writeChar(int), FilterOutputStream.out

writeUTF

```
public final void writeUTF(String str)
                    throws IOException
```

Writes a string to the underlying output stream using modified UTF-8 encoding in a machine-independent manner.

First, two bytes are written to the output stream as if by the `writeShort` method giving the number of bytes to follow. This value is the number of bytes actually written out, not the length of the string. Following the length, each character of the string is output, in sequence, using the modified UTF-8 encoding for the character. If no exception is thrown, the counter `written` is incremented by the total number of bytes written to the output stream. This will be at least two plus the length of `str`, and at most two plus thrice the length of `str`.

Specified by:

`writeUTF` in interface `DataOutput`

Parameters:

`str` - a string to be written.

Throws:

`UTFDataFormatException` - if the modified UTF-8 encoding of `str` would exceed 65535 bytes in length

`IOException` - if some other I/O error occurs.

See Also:

`writeChars(String)`

size

```
public final int size()
```

Returns the current value of the counter `written`, the number of bytes written to this data output stream so far. If the counter overflows, it will be wrapped to Integer.MAX_VALUE.

Returns:

the value of the `written` field.

See Also:

`written`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of

terms, workarounds, and working code examples. Other versions.

Java is a trademark or registered trademark of Oracle and/or its affiliates in the US and other countries.

Copyright © 1993, 2026, Oracle and/or its affiliates, 500 Oracle Parkway, Redwood Shores, CA 94065 USA.

All rights reserved. Use is subject to license terms and the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.